A Framework to Overcome the Dark Side of Generative Artificial Intelligence (GAI) Like ChatGPT in Social Media and Education

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***Abstract*— As the performance of generative artificial intelligence (GAI), such as ChatGPT, improves, content created by GAI will be distributed in the social media space, and knowledge and writings from unknown sources will be disseminated and reproduced. Now that GAI is becoming widespread, it is necessary to distinguish GAI from human intelligence, which constitutes knowledge. The data, information, knowledge, and work (DIKW) hierarchy is a useful framework for teaching and for checking metacognitive and explainable artificial intelligence (XAI) literacy. There are two types of collaboration between GAI and human intelligence: a combined intelligence model and a parallel intelligence model. The combined intelligence model is a method of using GAI for creating works by collecting data, organizing information, and deriving knowledge from information. This model is suitable for GAI-assisted tasks (GAIATs). The parallel intelligence model is suitable for GAI-assisted learning (GAIAL); it is a method in which a person develops abilities by analyzing and comparing tasks created by GAI after going through the data-information-knowledge-work process. The zone of proximal development (ZPD) created by educational scaffolding is a quantitative framework that is appropriate for evaluating the effects of GAI. The ZPD generated by GAI that corresponds to scaffolding should be managed so as not to favor or disadvantage specific individuals.**

***Index Terms*— Algorithm, ChatGPT, data, information, knowledge, and work (DIKW) hierarchy, education, explainable artificial intelligence (XAI), GAI-assisted learning (GAIAL), GAI-assisted tasks (GAIATs), generative artificial intelligence (GAI), social media.**

I. Introduction

Artificial intelligence (AI) is widely used in social media, such as Facebook, Instagram, and YouTube, because it has the potential to enhance user experience, increase engagement, and deliver personalized content. AI algorithms analyze user behavior and preferences to recommend personalized content that is the most likely to be relevant and engaging. AI can also analyze social media posts and comments to determine a user's feelings about a particular brand or topic. This information can be used to make data-driven decisions about marketing strategies and customer service. Chatbots and generative AI (GAI) are AI-powered tools that can interact with users on social media platforms. They can answer questions, provide customer support, and even make purchases on behalf of the customer [1].

Chatbots are becoming increasingly popular on platforms, such as Messenger and Twitter. By recognizing and analyzing the images and videos shared on social media, AI can provide outstanding services, products to buy, or identifying brand logos in user-created content.

ChatGPT is GAI. Its performance has recently improved dramatically, which is raising concerns as well as interest. ChatGPT is a powerful tool for natural language processing and communication, and it has a wide range of potential applications in fields, such as education, customer service, and healthcare [2], [3], [4]. Using GAI like ChatGPT is like using an Iron Man suit from Marvel Comics and movies. Similar to strengthening with an Iron Man suit, the existing data can be augmented or improved with ChatGPT.

Just as an Iron Man suit is equipped with powerful weapons that can deal with various situations, ChatGPT can manage several requests at the same time. GAI becomes more powerful as it continuously learns while processing big data. By using GAI, humans can experience augmented intelligence.

Several potential risks are associated with the widespread use of GAI. By providing users with content containing misinformation or knowledge, GAI algorithms may amplify the misinformation and knowledge [5].

GAI-powered tools often collect extensive data about users, including their search queries, search history, and social media activities. When such user data are shared with third parties, it may raise serious concerns about privacy and data security [6].

GAI algorithms may be biased because of the data they are trained on, which can further perpetuate stereotypes and discrimination. GAI also has the potential to automate much of the mental work that is currently performed by humans, and this could displace white-collar jobs and lead to economic inequality [7].

The main works and contributions of this article are as follows.

1) This study provides a theoretical framework for comparing GAI and human intelligence. The revised data, information, knowledge, and work (DIKW) hierarchy, which is a modification of the existing DIKW, is a useful framework for identifying the processes by which generative intelligence evolves data into information and evolves information into knowledge.

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Fig. 1. Bloom's taxonomy (2001).

2) The six levels of thinking presented by Bloom's revised Taxonomy as modified is a useful framework for comparing the characteristics of GAI and human intelligence.
3) This study provides strategies and frameworks for using GAI in work and education. The ability to evaluate the data, information, and knowledge generated by humans and to compare it with the data, information, and knowledge generated by GAI is important to the collaboration between AI and human intelligence. Strategies and frameworks must also be able to evaluate the reliability and validity of data, information, and knowledge generated by GAI.
4) This study provides a quantitative framework for evaluating the effectiveness of GAI in the areas of educational scaffolding and the zone of proximal development (ZPD). The effect of GAI is divided into a strong effect model, medium effect model, and limited effect model, and measuring and analyzing the volume of the ZPD area is also a meaningful approach in policy decision-making.

II. Framework for Knowledge Construction

A knowledge construction framework is a model that describes how individuals or groups generate new knowledge or revise existing knowledge through learning, experience, and reflection. Bloom's revised taxonomy and the DIKW (data, information, knowledge, and wisdom) hierarchy are useful knowledge construction frameworks for analyzing how GAI creates knowledge.

A. Bloom's Taxonomy

Published in 1956 and revised in 2001, Bloom's taxonomy is a framework for classifying educational goals and objectives. It is hierarchical and classifies according to the characteristics of thinking skills and knowledge. The taxonomy of thinking skills consists of six hierarchical levels of cognitive abilities, ordered from lower order thinking skills (LOTSs) to higher order thinking skills (HOTSs) [8] (Fig. 1).

According to this hierarchy, one cannot understand unless one remembers, one cannot apply without understanding, and one cannot analyze without applying. Similarly, one can evaluate only when one goes beyond the level of analysis, and one can create new things only when one reaches the level of evaluation.

The taxonomy is usually represented as a pyramid, with the lower level thinking skills forming the base and the higher level thinking skills building on top of it. Bloom's taxonomy has been widely used in education as a design tool for the development and assessment of learning objectives.

The verbs (remember, understand, apply, analyze, evaluate, and create) generally refer to the intended cognitive process. The objectives (factual knowledge, conceptual knowledge, procedural knowledge, and metacognitive knowledge) generally describe the knowledge that students are expected to acquire or construct [8].

Factual knowledge is knowledge that is fundamental to a specific field. It refers to essential facts, terms, details, or elements that must be known in order for a discipline to be understood or for a problem to be solved in that field. Conceptual knowledge is knowledge about classifications, principles, generalizations, theories, models, or structures

TABLE I
CHARACTERISTICS OF THE FOUR TYPES OF KNOWLEDGE

Knowledge	Characteristic of knowledge	Level
Metacognitive knowledge	·Strategic knowledge ·Cognitive tasks, including appropriate contextual and conditional knowledge ·Self-knowledge	Abstract knowledge
Procedural knowledge	·Subject-specific skills and algorithms ·Subject-specific techniques and methods ·Criteria for determining when to use appropriate procedures	
Conceptual knowledge	·Classifications and categories ·Principles and generalizations ·Theories, models, and structures	
Factual knowledge	·Knowledge of terminology ·Knowledge of specific details and elements	Concrete knowledge

TABLE II
BLOOM'S REVISED TAXONOMY SYSTEM
(THINKING LEVEL × KNOWLEDGE LEVEL)

Thinking skills level	Knowledge level			
	Factual knowledge	Conceptual knowledge	Procedural knowledge	Metacognitive knowledge
Create	Generate	Assemble	Design	Create
Evaluate	Check	Determine	Judge	Reflect
Analyze	Select	Differentiate	Integrate	Deconstruct
Apply	Respond	Provide	Carry out	Use
Understand	Summarize	Classify	Clarify	Predict
Remember	List	Recognize	Recall	Identify

related to a particular discipline. Procedural knowledge is information or knowledge that helps one perform tasks that are specific to a discipline, subject, or area of study. It may also be an investigation method, an extremely specific or finite technique, an algorithm, a technique, or a specific methodology.

Metacognitive knowledge is knowledge about one's own cognition and about oneself in relation to various subject matters [8]. Table I summarizes the characteristics of the four types of knowledge.

As shown in Table II, 24 behavioral verbs are produced by combining the six levels of thinking skills and four levels of knowledge. Of course, the behavioral verbs at the highest level, "generate," "assemble," "design," and "create," were thought to be behaviors that only humans could do. However, now that AI is able to think like that, it is necessary to clearly distinguish the subject of thinking.

B. DIKW (Data, Information, Knowledge, and Wisdom) Hierarchy

The DIKW hierarchy is a model that defines data, information, knowledge, and wisdom and it discusses their interrelationships (Fig. 2). Data are created through research, generation, gathering, and discovery. Information is data with

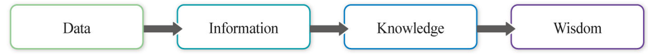

Fig. 2. DIKW hierarchy model.

Fig. 3. DIKIW hierarchy model.

context added to it; we transform data into information by organizing the data, so that conclusions can easily be drawn from it. Data visualization is a prime example of information.

The difference between information and knowledge can be identified as lower and upper concepts, such as the relationship between data and information. Information is data arranged in a meaningful way, and knowledge means applying and productively using information [9]. While information has a passive character and is received from the outside, knowledge involves an active process, in which a person who has received information processes and judges it by his or her own initiative. In other words, when information obtained through a selection process is acted on through the process of reconstruction, accumulation, and judgment, it is reborn as knowledge.

Knowledge is the result of looking at information from multiple perspectives. Information is static, but knowledge is dynamic because it lives within us. Wisdom is the ultimate level of enlightenment beyond knowledge. As with knowledge, wisdom operates within us [10].

Although the DIKW hierarchical structure has been criticized as methodologically undesirable because defining data, information, and knowledge incurs a logical fallacy known as circular definition [10], [11], the structure provides a framework that can be used as a management standard in the information sciences.

This hierarchical model is mainly used for data-information-knowledge level, but the wisdom level is rarely discussed. This is because the relationship between knowledge and wisdom is heterogeneous and abstract, unlike the relationship between data and information or that between information and knowledge.

DIKIW hierarchy is a model that data, information, knowledge, intelligence, and wisdom is a revised DIKW hierarchy model. Intelligence plays an important role in the relationship between knowledge and wisdom in the DIKW hierarchy model [10] (Fig. 3).

However, because intelligence is required even for the process of developing data into information and developing information into knowledge, a model that explains that intelligence is necessary only between knowledge and wisdom is inevitably weak in explanatory power. Efforts to eliminate the ambiguity of knowledge and wisdom continue, but no solution has been obtained yet [12].

III. EDUCATIONAL FUNCTIONS AND EFFECTS OF GAI

A. GAI as Educational Scaffolding

Scaffolding is a temporary structure that is installed for workers at construction sites. After cognitive psychologist

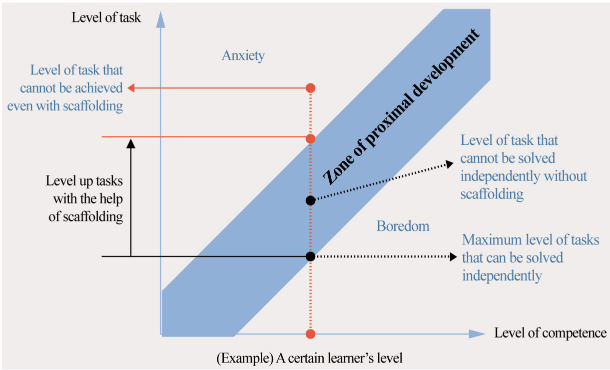

Fig. 4. ZPD formed by educational scaffolding.

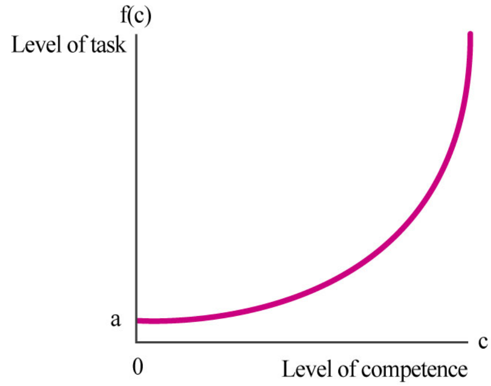

Fig. 5. Powerful effect model for GAI.

Lev Semenovich Vygotsky (1896–1934) used this concept, it became a widely used concept in education [13]. In the field of pedagogy, providing scaffolding refers to the provision of assistance by teachers or peers, so that students or beginners can perform a given learning task well.

Scaffolding can be divided into physical and abstract. An adult teaching a child how to ride a bicycle is abstract scaffolding, and the two small training wheels attached to the bicycle correspond to physical scaffolding. Just as temporary scaffolding is removed after construction is finished, the help provided in the early stages of learning should be gradually reduced as the learning progresses to allow the learners to perform a task by themselves. If child has become accustomed to riding a two-wheeled bicycle, adult will first remove one training wheel, and if child no longer needs the training wheels, adult will remove the other training wheel.

The level of competence in which a learner can perform with the help of scaffolding is called the ZPD. The following graph expresses the relationship between the task level and an individual's ability [14] (Fig. 4).

The x-axis is the learner's competence level, and the y-axis is the level of tasks that are achievable. Generally, the level of competence and the level of achievable tasks are proportional, so the relationship between the independent variable and the dependent variable can be simply expressed as a linear function, that is, a straight line.

If the level of the task is lower than the learner's ability, this learning situation becomes the area under the linear function line, which is the area where the learner gets bored. Conversely, if the level of the task is higher than the learner's ability, learning situation corresponds to the region above the linear function line, and the learner enters a state of anxiety.

When scaffolding is used, the linear function line shifts upward. In this case, the area between the first straight line and the straight line that is shifted upward corresponds to the area where the learner can develop with outside help. This is the ZPD that Vygotsky emphasized.

For example, the Y value where the straight line that is shifted upward crosses the first straight line is the level of the task that the learner can solve alone. If abstract scaffolding (help from teachers or peers) or physical scaffolding (learning tools and learning infrastructure) is provided, the level of tasks that learners can perform increases. That is, the level of tasks that learners can perform can be raised to the second straight line. If one is provided with adequate scaffolding, one can tackle higher level tasks that would be possible alone.

GAI, which ChatGPT is an example of, can be used as an overwhelming educational scaffold. The ZPD formed when GAI is used as an educational scaffolding is also incomparably wider than any other educational scaffolding it can elevate ordinary people to semi-professional levels.

B. Effects of GAI

The level of task outcomes that can be achieved using GAI can be assumed by three effect models. A powerful effect model assumes that if the user's ability is high, the task level can be increased almost infinitely. In contrast, limited effect models assume that it is impossible to increase the level of the task beyond a certain level. A medium effect model is an ideal model that assumes that the level of the task can be increased to a certain level, regardless of the user's competence level.

1) Powerful Effect Model: A powerful effect model for GAI can be expressed as an exponential growth model. If the level of competence (C) is specified on the x-axis and the task level (T_1) is set on the y-axis, it can be estimated by an exponential function such as follows:

$$T_1 = f(C) = ae^{\alpha C} \tag{1}$$

where T is the level of the task, C is the level of competence, a is the value of the initial task level, α is the coefficient of C (the rate of increase in task level with competence level), and e is the natural constant (approximately 2.718).

GAI's powerful effect model, which is expressed as an exponential function, shows that the level of tasks achievable with the use of GAI as scaffolding increases exponentially as an individual's competence level increases (Fig. 5).

As shown in Fig. 6, which shows the ZPD created by the powerful effect model, a very wide ZPD corresponding to infinity can be secured by using GAI as scaffolding to transform the initial linear function to an exponential function. This powerful effect model suggests that, contrary to some expectations, GAI is not wiping out all knowledge workers but rather is presenting a better opportunity for the more capable ones.

According to this model, people with high levels of ability can increase their job performance levels to a degree that

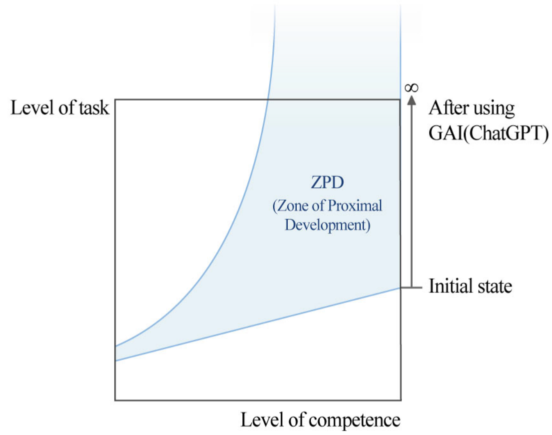

Fig. 6. ZPD of powerful effect model.

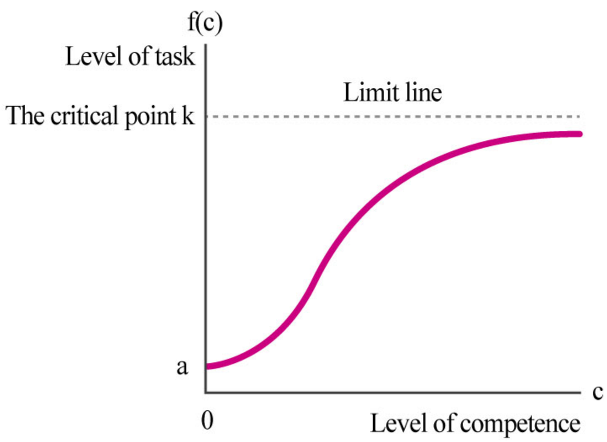

Fig. 7. Limited effect model for GAI.

people with low levels of ability can never match, resulting in the so-called "the rich get richer, the poor get poorer" phenomenon.

2) Limited Effect Model: A limited effect model for GAI can be expressed as a logistic function model. If the level of competence (C) is specified on the x-axis and the task level (T_2) is specified on the y-axis, it can be estimated by a logistic function such as follows:

$$T_2 = f(C) = a + (k - a)\frac{k}{1 + e^{-\alpha C}} \tag{2}$$

where T is the level of the task, C is the level of competence, a is the value of the initial task level, α is the coefficient of C (the rate of increase in task level with competence level), k is the critical point of task level, and e is the natural constant (approximately 2.718).

The logistic function generated by the limited effect model is a graph that is similar to an S-curve, which starts from a low value and gradually increases to a maximum point and then reaches a plateau with no further increase. The higher a person's competency level, the easier it is to perform more complex tasks. However, if the task difficulty is above the threshold, at some point, it stagnates at some point or even declines as it exceeds the level of human capabilities. This point is the saturation point of the logistic function (Fig. 7).

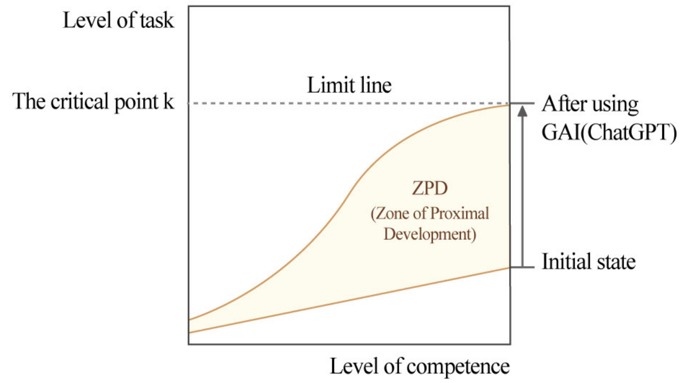

Fig. 8. ZPD of the limited effect model.

GAI's limited effect model, expressed as a logistic function, shows that as an individual's ability level rises from low to intermediate, the level of the tasks that can be achieved using GAI also increases but does not increase beyond a certain threshold. In other words, the model indicates that we face a "ceiling effect." Therefore, the output produced by GAI cannot exceed the output produced by human ability.

According to Fig. 8, which shows the ZPD generated by the limited effect model, a wide ZPD can be secured at the intermediate level or higher by converting the initial linear function to a logistic function. This model supports the argument that GAI cannot completely replace knowledge workers.

Modeling the relationship between competence level and task level using a logistic function in this way provides quantitative insight into the optimal competency level needed by workers to effectively perform tasks utilizing GAI. This allows us to establish quantitative criteria for designing human resource training programs.

3) Moderate Effect Model: A moderate effect model for GAI can be expressed as a linear model. If the level of competence (C) is specified on the x-axis and the task level (T_3) is specified on the y-axis, it can be estimated by a linear function such as follows:

$$T_3 = f(C) = a + \alpha C \tag{3}$$

where T is the level of the task, C is the level of competence, a is the value of the initial task level, and α is the coefficient of C (the rate of increase in the task level with competence level).

This model makes the simple assumption that the level of tasks that can be achieved using GAI is a linear function of competence level from a low level of competence to a high level of competence. According to the moderate effect model for GAI, the level of task that can be achieved using QAI such as ChatGPT is directly proportional to the user's competence (Fig. 9).

The linear assumption that this model presupposes is educationally desirable. In previous studies that model the effect of scaffolding quantitatively, expressing ZPD as a space created by a shift in a linear function is also because it makes an educationally desirable assumption [15].

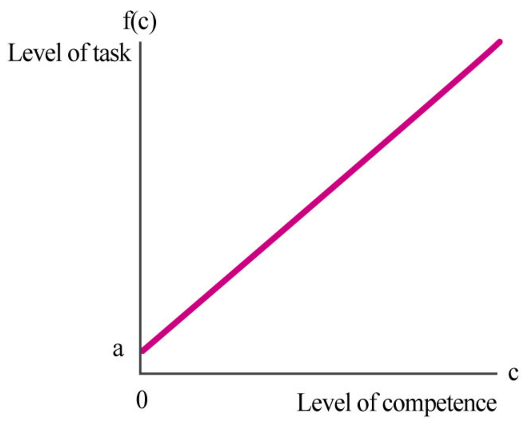

Fig. 9. Moderate effect model for GAI.

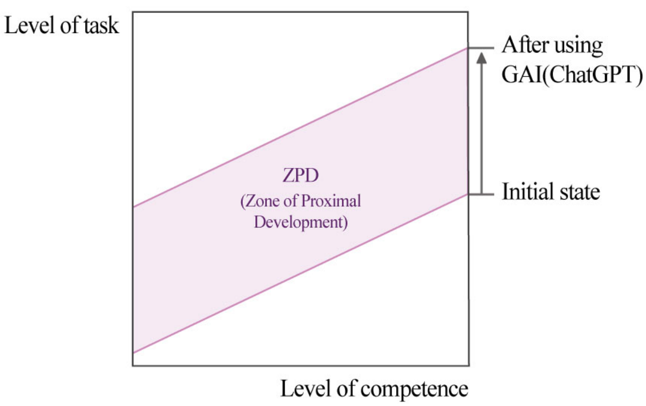

Fig. 10. ZPD of the moderate effect model.

As shown in Fig. 10, which expresses the ZPD generated by the moderate effect model, a relatively wide ZPD than for other models can be established by parallel shifting the initial linear function by GAI as a stepping stone. As such, the moderate effect model suggests that GAI, provided appropriately, can greatly benefit users of all skill levels.

However, this linear model involves an overly romantic assumption, and in practice, the relationship between ability level and task level is rarely directly proportional. Therefore, a linear model describes a desirable and ideal state to be improved through education rather than a phenomenon.

IV. Framework for GAI-Assisted Task

A. Revised DIKW Hierarchy and the Work Society

An information-oriented society was formed the wide-ranging social and organizational changes that resulted from the information and communications revolution. A knowledge society is a society in which information is combined with human creativity and in which knowledge develops as a new factor of production [9]. In a society where humans and AI collaborate to create work, the term "work" is needed instead of the ambiguous term "wisdom." If the stage after the generation of knowledge (the postknowledge stage) is defined as the creation of work, the concept of "the work society" can be created on an equal footing with "the information-oriented society" and "the knowledge society." The work society is a society that gives value to works created using the existing knowledge rather than knowledge itself, and it is a concept that is suited to the era of GAI (Fig. 11).

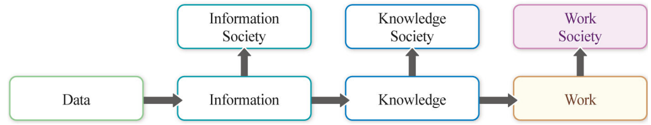

Fig. 11. Revised DIKW hierarchy and the work society.

B. Human-GAI Collaboration Model (Combined Intelligence Model)

GAI is not an intelligence that has already been completed but a potential intelligence that develops through interaction with people [16], [17]. Using GAI is like interacting with other people. GAI offers the advantage of reducing mistakes and routine tasks while interacting with others and improving what people already know.

GAI should be applied as a different model in the educational field and in the workplace. GAI-assisted tasks (GAIATs) are a way to refer to data, information, and knowledge generated by GAI, such as ChatGPT as humans construct data, information, and knowledge (Fig. 12). This can be expressed as a combined intelligence model or a human-GAI collaboration model. According to this model, humans can use GAI as an assistant or a colleague, and they can achieve outstanding results that cannot be compared to working alone. If teaching and learning are designed based on this model in the field of education, it will eventually learn how to work and live with AI.

V. Framework for GAI-Assisted Learning

A. Educational Application of the Revised DIKW Hierarchy

Exploration activities or project-based learning are effective for inducing activities that are suited to the student's level by organizing them into a hierarchy of DIKWs according to the student's level of thinking. If the student's level or time is insufficient, it ends at the level of data collection or information creation, but if the student has sufficient time, it can develop to the level of knowledge production and work production.

The data levels can be classified as obtaining raw materials through gathering and discovery and as measured values obtained directly through experiments. The former is found data (FD), and the latter is generated data (GD). The information levels that are created by processing data can be classified as found information (FI) or generated information (GI).

The knowledge obtained from processing information, on the other hand, can be classified as found knowledge (FK) or generated knowledge (GK). Distinguishing between FK and GK is intended to separate quotation and argumentation. The work is something that is obtained by converging the FK and GK through presentation and discussion and by developing a work. Some representative forms of works are the portfolio, research paper, thesis, and video clip. The results at the data level are survey reports, and the results at the information level are inquiry reports.

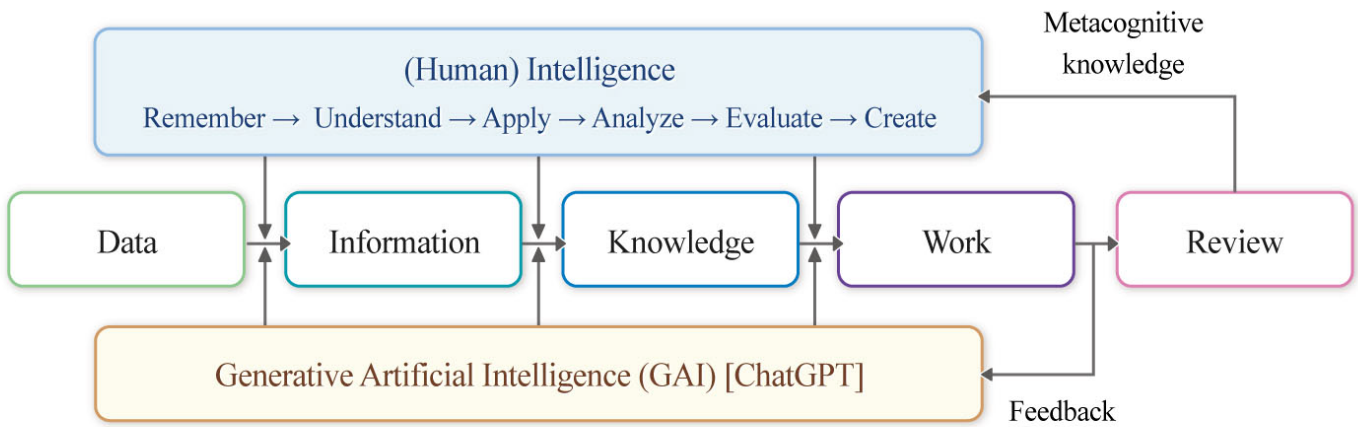

Fig. 12. Human-GAI collaboration model (combined intelligence mode).

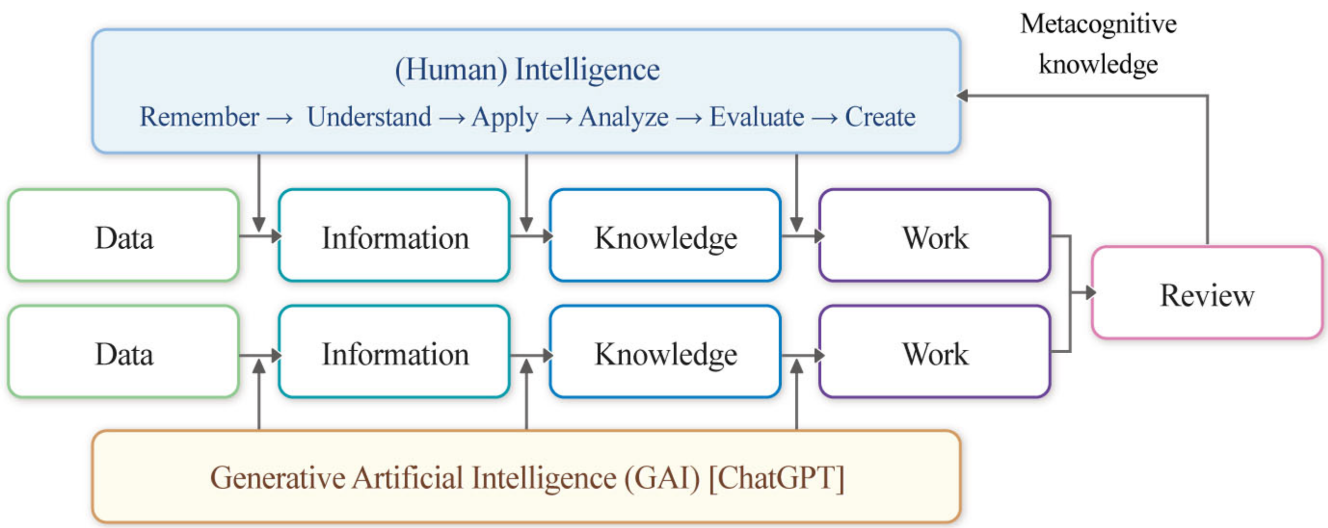

Fig. 13. Human-GAI parallel processing model (parallel intelligence model).

Explaining the self-generated GD, GI, GK, and GW while attributing the sources of FD, FI, FK, and FW is a useful strategy for teaching research ethics and research paper writing. As shown, by separating the given from the self-created, objectivism and constructivism may be compromised.

Through projects, students can develop the competency to develop task-related materials to the level of information and knowledge. They also develop the competency to differentiate between what is given and what is created and to express both as works. During this process, learners will learn what to cite and what to claim, so they naturally develop research ethics as well as experience communication conflicts and develop the ability to overcome trial and error.

B. Human-GAI Parallel Processing Model (Parallel Intelligence Model)

1) XAI Literacy: With the release of ChatGPT 4.0 in 2022, the power of generative AI suddenly became powerful. Therefore, it is important to distinguish when to use GAI aggressively and when to use it cautiously [18].

AI-assisted learning is a new type of technology based on big data analysis, deep learning, and neural networks that provide appropriate recommendations and personalized solutions for learners' individual situations and needs and thereby achieving precision teaching.

GAI-assisted learning (GAIAL) is a strategy for activating metacognition by comparing learners with works created by GAI after developing them into data-information-knowledge by themselves and completing them as works.

This metacognition is a look back at the processes that GAI carries out to achieve its goals, so it eventually corresponds to explanatory artificial intelligence (XAI) literacy.

2) GAI-Assisted Learning: If the objective combined intelligence model discussed above was to construct knowledge while referencing AI, the parallel processing model presented in Fig. 13 is a method, in which humans construct knowledge first and then compare that knowledge with the knowledge created by AI.

If the ability to judge human-created knowledge as one's own through reflective thinking is metacognitive, then understanding the algorithm by which AI produces knowledge is the process of developing XAI literacy. The DIKW hierarchy

TABLE III
DIKW HIERARCHY (FOUND AND GENERATED X)

Level	Characteristic of outputs		Interpretation
Data	FD	Found Data	Material reconstructed from material created by others
	GD	Generated Data	Data obtained through experiments or surveys
Information	FI	Found information	Information revealed by others
	GI	Generated information	Analyzing data and assigning meaning to it
Knowledge	FK	Found knowledge	Knowledge revealed by others
	GK	Generated knowledge	Generating something subjective by analyzing information
Work	FW	Found work	Work revealed by others
	GW	Generated wok	Creating something subjective by analyzing knowledge

model is a favorable framework for triggering and affirming local cognition and XAI literacy (Table III).

XAI literacy can be developed by comparing and evaluating human-made data, information, and knowledge with GAI-created data, information, and knowledge. If one relies too heavily on GAI from the start, one will not develop the ability to evaluate the reliability and validity of the data, information, and knowledge generated by GAI. Eventually, this prevents the final task performance from exceeding a certain level [19], [20].

If students are overly reliant on GAI, such as ChatGPT, before they enter society, they will be reduced to text editors rather than text producers. Therefore, in order to design GAI-assisted learning, it is necessary to clearly distinguish between GAIAL and GAIAT, as summarized in Table IV.

In an era of data, information, and knowledge overload, it is even more important to develop the ability to assess their reliability and validity. GAIAL requires the following guidelines.

1) Understand the nature of data, information, knowledge, and work.
2) Be able to distinguish between data, information, and knowledge generated by learners and data, information, and knowledge generated by GAI (ChatGPT).
3) Be able to be compare and evaluate data, information, and knowledge generated by learners and data, information, and knowledge generated by GAI (ChatGPT).
4) Based on the data, information, and knowledge generated by GAI (ChatGPT), be able to identify problems that arise when data, information, and knowledge accumulate.
5) Be able to evaluate the reliability and validity of data, information, and knowledge generated by GAI (ChatGPT).

TABLE IV
GAIAL AND GAIAT

Strategy	GAI-assisted learning	GAI-assisted tasks
Abbreviation	GAIAL	GAIAT
Circumstances	GAI in the classroom	GAI in the workplace
Concept	A learning method that compares and analyzes data, information, and knowledge created by GAIs such as ChatGPT after humans produce data, information, and knowledge on their own	The working method for constructing the data, information and knowledge by referring to data, information and knowledge created by ChatGPT
Role	· Infrastructure for teaching and learning · Educational scaffolding · Desirable expansion of ZPD	· Competent secretary or colleague · Overwhelming scaffolding · Dramatic expansion of ZPD
Effect	· Moderate effect model	Powerful effect model
Principle	· Parallel intelligence model	· Combined intelligence model

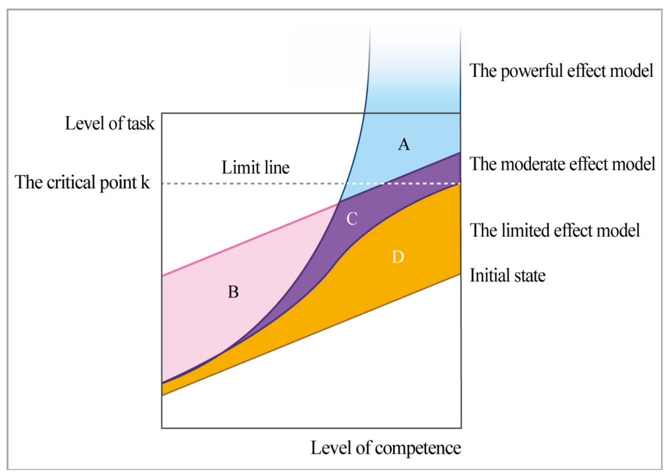

Fig. 14. Comparison of ZPD for three effect models.

6) Understand the difference between the method for comparing data, information, and knowledge with those first created by humans and then generated by GAI (ChatGPT) and the method for constructing knowledge by referring to the data, information, and knowledge generated by AI.
7) Understand that if humans are overly reliant on GAI (ChatGPT), the development of their cognitive thinking and problem-solving competence may be limited.

VI. CONCLUSION

This article proposed three frameworks that are necessary for the era in which GAI and humans coexist. First, the DIKW hierarchical structure is a useful framework comparing GAI and the human knowledge creation process. Second, the six levels of thinking and four levels of knowledge presented by Bloom's revised taxonomy are useful frameworks for comparing the characteristics of GAI and human intelligence. Third, the ZPD created by educational scaffolding is an appropriate quantitative framework for evaluating the effects of GAI.

The effects of GAIAT can be described by powerful effect models. This is because when GAI is used for work, the greater the user's ability, the higher the ZPD can be. The effect of GAIAL can be described by a limited-effects model. The higher the learner's ability, the higher the ZPD, but the limits are clear. Therefore, the most desirable state is the medium-effect model state. This is because in terms of human resource management, increasing the achievement of the middle and lower levels is important, whether for education or business purposes. Moreover, a person with a high level of ability should be able to overcome their limitations.

When GAI is used for training, scaffolding must be set up, so that a ZPD can be formed for the medium-effect model state. Area A in Fig. 14 is the area of inequality due to GAI. Area B is a scaffolding to be provided to learners below the intermediate level, and area C should be secured, so that learners above the intermediate level can overcome their limitations. Without preparation for GAIAL, the area of proximal development that GAI can produce is limited to Area D.

This article is a proposal of an educational framework that is needed for the era in which GAI and humans coexist. A follow-up study will develop a detailed protocol for a quantitative comparative procedure by measuring the ZPD and presenting a specific algorithm that can be readily be used as an evaluation tool for the ZPD.

In conclusion, the ZPD generated by GAI that corresponds to scaffolding should be managed so as not to favor or disadvantage specific individuals.

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